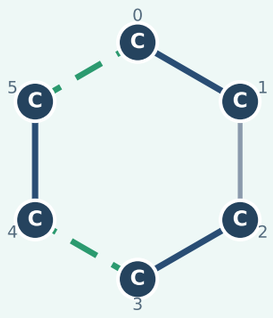
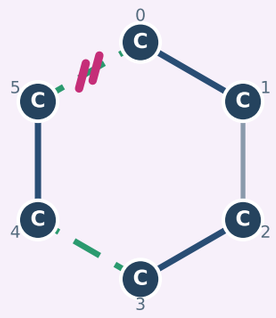
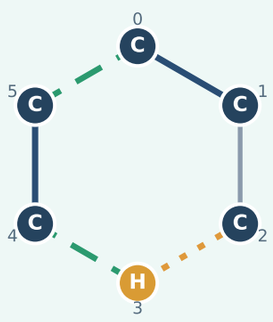
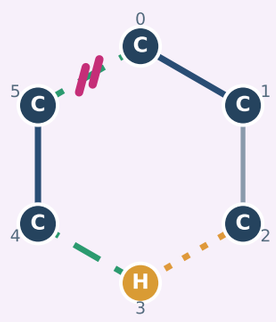
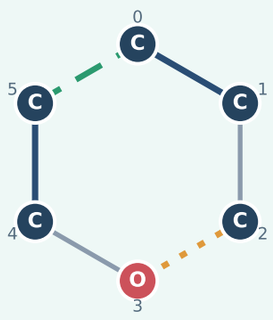
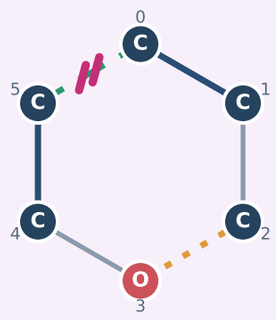
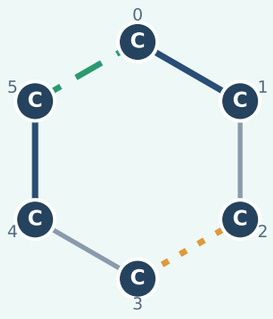
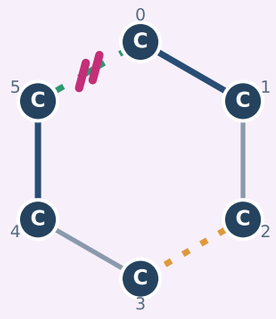
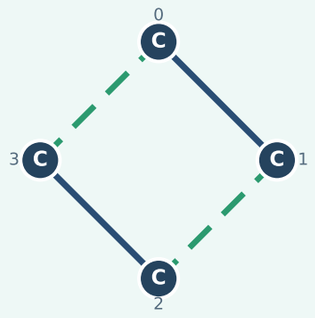
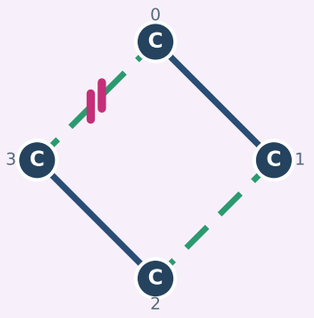
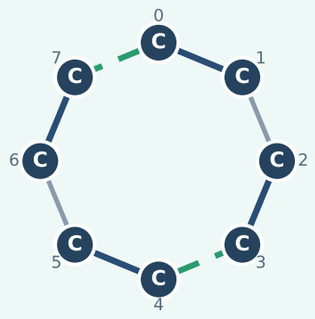
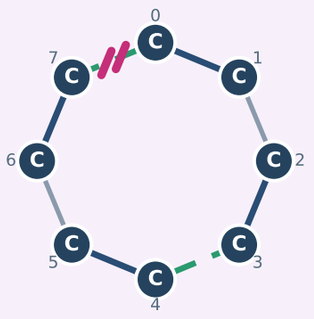
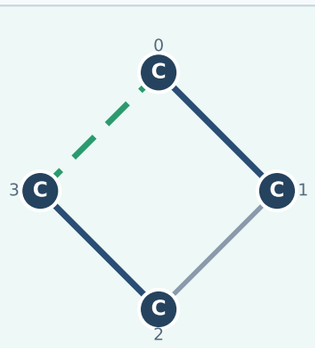
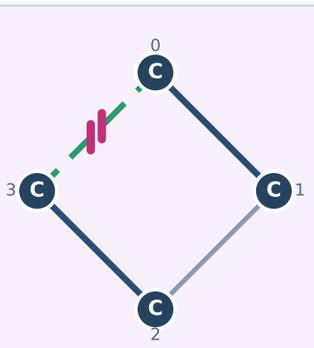
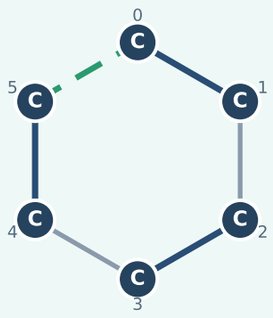
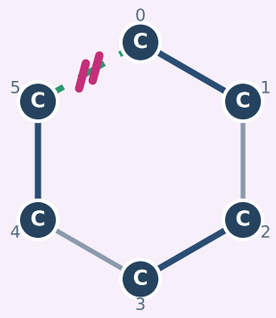
How the approach changes the orbital circuit

Each row holds the electron count fixed and changes the proposed orbital phase topology.

	HÜCKEL • no phase inversion • Φ = +1	MÖBIUS • one phase inversion • Φ = −1
Diels–Alder [4+2] 6 active e⁻ • 4π + 2π	 suprafacial / suprafacial **ALLOWED** loop phase +1	 suprafacial / antarafacial **FORBIDDEN** loop phase -1
Alder–ene 6 active e⁻ • 2π + 2π + C–H σ	 all suprafacial **ALLOWED** loop phase +1	 one antarafacial component **FORBIDDEN** loop phase -1
Claisen [3,3] 6 active e⁻ • 2π + 2π + C–O σ	 suprafacial / suprafacial **ALLOWED** loop phase +1	 suprafacial / antarafacial **FORBIDDEN** loop phase -1
Cope [3,3] 6 active e⁻ • 2π + 2π + C–C σ	 suprafacial / suprafacial **ALLOWED** loop phase +1	 suprafacial / antarafacial **FORBIDDEN** loop phase -1
[2+2] cycloaddition 4 active e⁻ • 2π + 2π	 suprafacial / suprafacial **FORBIDDEN** loop phase +1	 suprafacial / antarafacial **ALLOWED** loop phase -1
[4+4] cycloaddition 8 active e⁻ • 4π + 4π	 suprafacial / suprafacial **FORBIDDEN** loop phase +1	 suprafacial / antarafacial **ALLOWED** loop phase -1
4π electrocyclic closure 4 active e⁻ • 4π	 disrotatory **FORBIDDEN** loop phase +1	 conrotatory **ALLOWED** loop phase -1
6π electrocyclic closure 6 active e⁻ • 6π	 disrotatory **ALLOWED** loop phase +1	 conrotatory **FORBIDDEN** loop phase -1

π bond · conjugated link · forming bond · breaking bond · one orbital phase inversion

Green means formally thermally allowed; red means symmetry-forbidden for that specified concerted approach.

A formally allowed antarafacial route may still be geometrically inaccessible. No 3D transition-state geometry was optimized.